

Nuojunxi (Steven) Zhang

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Education

- University of South Florida**, Ph.D. in Computer Science & Engineering Aug 2026 – Present
- Bellini College of Artificial Intelligence, Cybersecurity and Computing
 - Advisor: Prof. Yi Sheng
- Kean University**, B.S. in Computer Science Sept 2022 – May 2026
- Degree completed at Kean University, Union, NJ
 - **Coursework:** Data Structures and Algorithms, Computer Architecture, Object-Oriented Programming
 - Dean's Honor List, Kean University (term GPA 3.92/4.00, Aug 2024 – Dec 2025)

Technologies

Languages: Java, Python, C#, SQL, MATLAB
Tools & Frameworks: PyTorch, TensorFlow, OpenCV, Hugging Face, Git, Linux, Docker, Unity, Object-Oriented Design, System Testing, Technical Writing

Publication

- Segmenting What Matters: A Dual-Stage Active Learning Framework for Weakly Supervised Breast Ultrasound Segmentation** October 2025
Nuojunxi Zhang, Meng Xu, Guanchao Tong, Kuan Huang
IEEE International Conference on Bioinformatics and Biomedicine (BIBM) 2025
- Segmenting with Reliable Guidance: Dual-Branch AMR and CAM-Guided Learning for Weakly Supervised Breast Ultrasound Segmentation**
Nuojunxi Zhang, Kuan Huang
Manuscript submitted to IEEE Journal of Biomedical and Health Informatics (JBHI)

Experience

- Graduate Research Assistant**, YES Lab, University of South Florida Aug 2026 – Present
- Supervised by Prof. Yi Sheng, working on X-ray CT reconstruction and detector calibration
 - First-author abstract under review at SPIE Medical Imaging 2027
- Graduate Instructional Assistant**, Bellini College of AI, Cybersecurity and Computing, University of South Florida Aug 2026 – Present
- Instructional assistant for a course on incident response methods
 - Supporting labs and hands-on exercises, grading, and office hours
- Research Assistant (NSF-Funded)**, AI4Healthcare Lab, Kean University June 2025 – Sept 2025
- Appointed under NSF funding, supervised by Prof. Kuan Huang
 - Contributing to a journal-targeted research on ultrasound-based medical image segmentation
 - Responsible for algorithm development, experimental design, and manuscript preparation
- Research Assistant (NSF-Funded)**, AI4Healthcare Lab, Kean University Dec 2024 – May 2025
- Designed a dual-stage active learning framework for weakly supervised tumor segmentation using image-level ultrasound labels
 - Developed HSV-based CAM filtering and SAM-guided pseudo-label refinement to enhance mask quality
 - Integrated a Mean Teacher segmentation model with iterative uncertainty sampling, achieving 68.25% IoU and 79.39% DSC on the BUSI dataset
 - Conducted end-to-end project design, experimentation, and publication as an NSF-funded research assistant
 - **First-author paper accepted at IEEE BIBM 2025** (acceptance rate: 19.8%)

- Research Participant**, Medfusion Fake Image Generator, Kean University Sept 2024 – Dec 2024
- Developed a synthetic medical image pipeline using Variational Autoencoders (VAE) and Diffusion Models to augment breast ultrasound datasets for weakly supervised segmentation
 - Training a latent VAE embedder and conditional diffusion model on BUSI dataset
 - Designed experiments using diffusion-generated images and weak labels to evaluate their impact on downstream segmentation performance
- Research Assistant**, Vision-AI Lab, Wenzhou-Kean University Sept 2023 – May 2024
- Joined Prof. Gupta's Vision-AI Lab to support projects in computer vision and medical imaging
 - Participated in a SPF-funded initiative focused on raw image denoising using Bayer pattern modeling
 - Worked on latent space extensions and codebase adaptation based on a prior CVPR 2022 Workshop project developed by a former lab member under the same advisor
- Vice President**, Wenzhou-Kean University Computer Club Jan 2024 – June 2024
- Managed and planned club operations, overseeing departmental organization and project planning
 - Led strategic initiatives that increased active membership and project participation by 30%
- College Assistant**, College of Science, Mathematics and Technology, Wenzhou-Kean University Aug 2022 – May 2024
- Assisted faculty with event promotion and day-to-day coordination of academic activities
 - Streamlined scheduling workflows, improving operational efficiency across departmental projects
- Research Assistant**, Parkinson's Rehabilitation System, Wenzhou Kean University Feb 2023 – June 2023
- Utilizing OpenCV and Mediapipe, I developed a program that captures patients' hand movements and provides real-time feedback for rehabilitation assessment
 - Earned the *Outstanding Presentation* award at WKU Research Day for demonstrating prototype results

Projects

- DSAL: Dual-Stage Active Learning for Medical Segmentation** *Segmentation, Medical Imaging*
- Proposed a dual-stage framework combining weakly supervised segmentation with active learning
 - Achieved 68.25% IoU and 79.39% DSC on BUSI dataset; reduced annotation costs.
 - Published as **First Author** paper at **IEEE BIBM 2025**.
- arXivPush: Automated arXiv Paper Summarization and Daily Digest System** *LLMs, NLP Pipeline*
- Designed an end-to-end system that automatically fetches, deduplicates, summarizes, and translates new arXiv papers into bilingual daily digests.
 - Integrated Discord API, Python automation, and LLM summarization (Ollama + Qwen2.5) for seamless academic community delivery.
 - Released as an open-source project on GitHub, earning **29+ stars**, and reaching over **6.8K views** and **1300+ engagements** across communities.
 - Companion iOS client currently in **TestFlight** beta.
- Medfusion-Fake-Image Generator (Extension Project)** *Diffusion Models, VAE, Medical Imaging*
- Extended the open-source Medfusion framework by integrating pathology-aware training for realistic, tumor-inclusive ultrasound image synthesis.
 - Implemented 512×512 high-resolution generation with distributed training and customizable configurations for medical AI research.
- DeepTeacher: AI-Powered Learning Assistant** *Multimodal LLM System, Vision-Language Models*
- Built a local privacy-preserving learning assistant combining LLaVA-Phi3 and DeepSeek-R1 for real-time screen understanding and feedback.
 - Enabled multimodal interaction with integrated image reasoning and adaptive dialogue generation.